

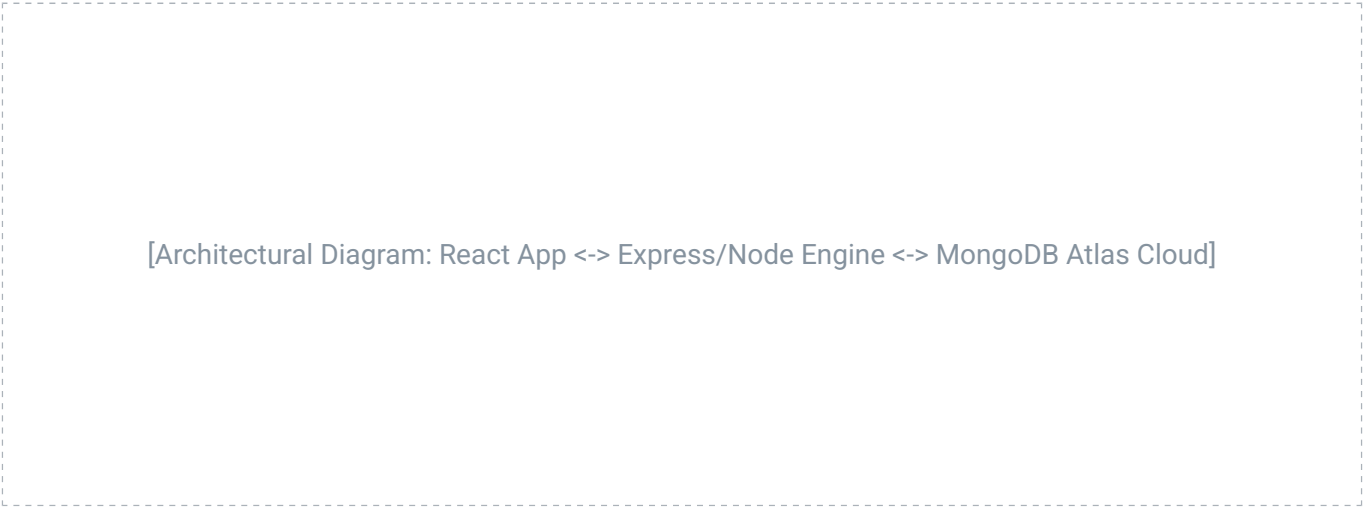
Project Design Phase-II

Technology Stack Infrastructure & Architecture

Submission Date	31 January 3035
Team ID	
Project Name	ShopEz
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Assessment Weight	4 Marks

Technical Solution Architecture

The system design provides a comprehensive roadmap of the ShopEz e-commerce ecosystem, illustrating how front-end components connect with backend logic and cloud data services. This deliverable encompasses the multi-tier interaction model between the client-side interface and the MERN stack utility blocks.



Documentation Standards:

- Map logical blocks to specific application processes.
- Clarify boundaries between local development and cloud integration.
- Detail external API touchpoints and third-party gateways.
- Specify storage layers and persistent data components.

Table-1: Component & Technology Integration

S.No	Component	Functional Description	Implementation Technology
1.	Interface Layer	Responsive user interaction via Web and Mobile browsers.	React.js, HTML5, Tailwind CSS
2.	Server Logic	Orchestrating API endpoints and business rules.	Node.js & Express.js Engine
3.	Auth Services	Secure login handlers and token generation logic.	JWT (JSON Web Tokens)
4.	Image Assets	Handling product imagery and static multimedia storage.	Cloudinary / Local Filesystem
5.	Primary Database	Structural data storage for users, orders, and products.	NoSQL (MongoDB)
6.	Cloud Storage	Remote database hosting for global accessibility.	MongoDB Atlas Cloud
7.	External API-1	Integration for processing secure online transactions.	Stripe / Payment Gateway API
8.	External API-2	Email automation for order and account verification.	Nodemailer / SendGrid
9.	Infrastructural Host	Production deployment and environment management.	Heroku / Vercel / AWS

Table-2: System Attributes & Performance Characteristics

S.No	Characteristic	Architectural Rationale	Applied Technology
1.	Modular Frameworks	Leveraging pre-built open-source modules for speed.	MERN Stack Ecosystem
2.	Data Security	Implementing end-to-end protection and encryption.	Bcrypt, HTTPS, JWT
3.	Scalability	Designing for future traffic expansion via 3-tier models.	Stateless API Architecture
4.	Platform Uptime	Ensuring continuous availability through cloud hosting.	MongoDB Atlas, Heroku
5.	Response Speed	Optimizing latency with efficient query handling.	Index-based DB Queries